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UNIVERSITY OF GYŐR

Bio-Inspired Routing Optimization for Complex Logistics Problems

Boldizsár Tüű-Szabó

Széchenyi István University, Department of AI and HPC, Győr, Hungary

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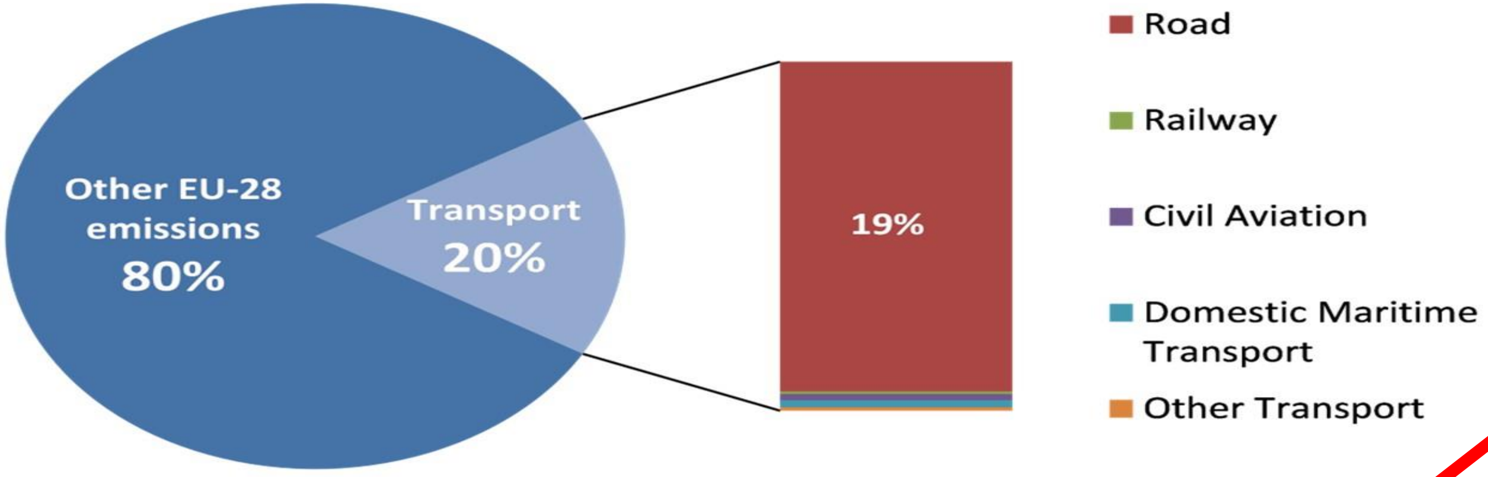
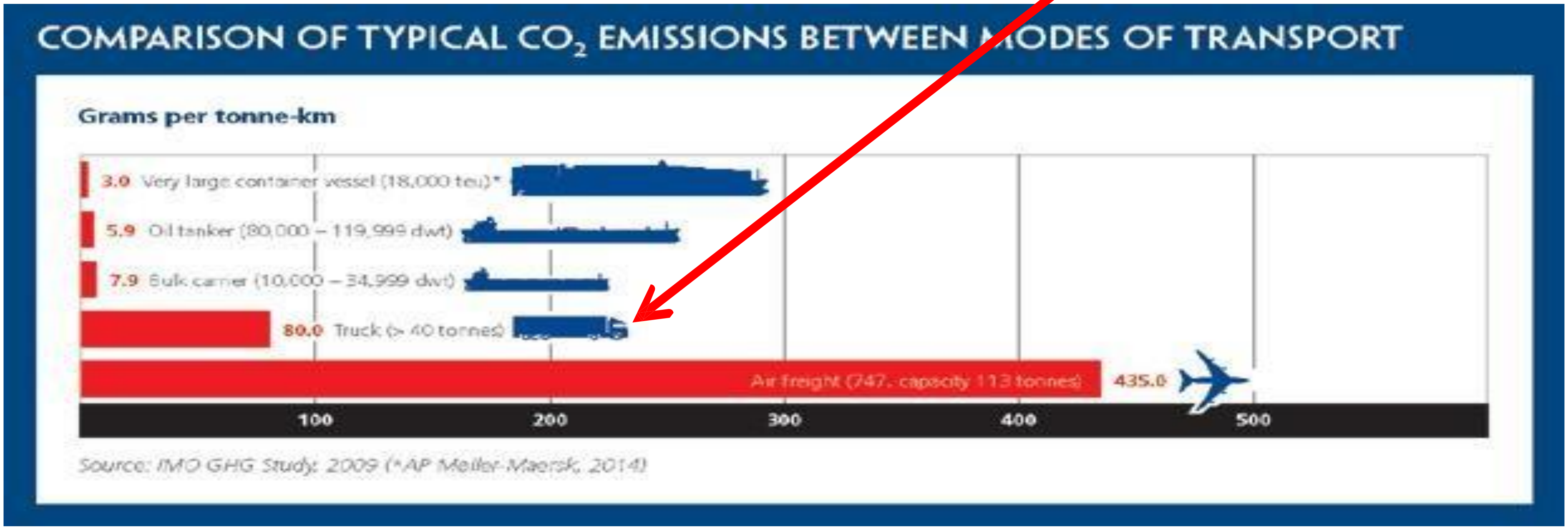
Content

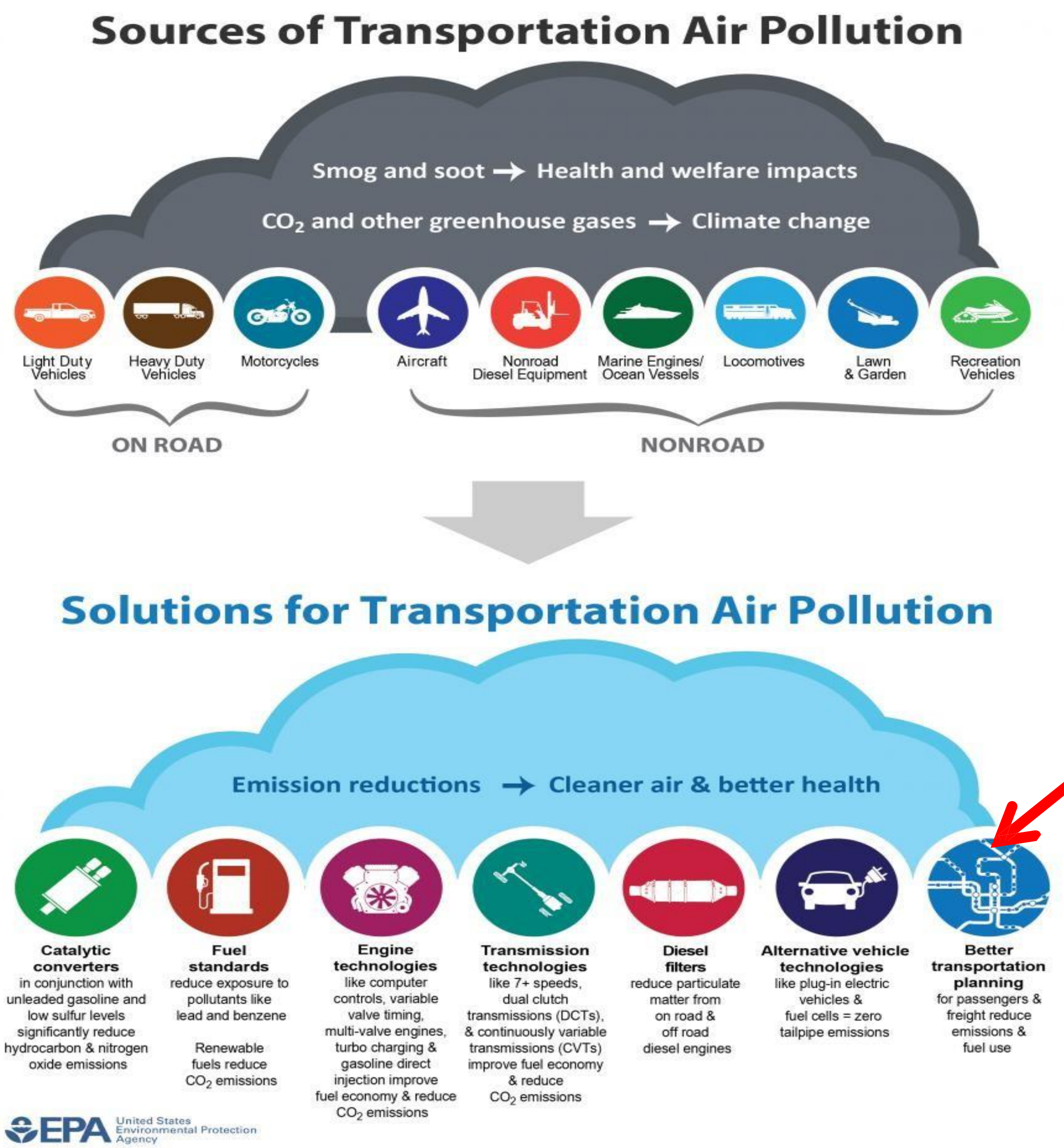
- **Urban Logistics & Challenges**
  - *Sustainability, traffic congestion, and strict delivery time windows*
- **Uncertainty in Transportation Networks**
  - *Modeling non-deterministic travel times with Fuzzy & Intuitionistic Fuzzy sets*
- **Soft Computing Approach (DBMEA)**
  - *Bio-inspired metaheuristics for complex routing problems*
- **Experimental Results & Applications**
  - *TSP, Time-dependent TSP models*
- **Conclusions**

AIR POLLUTION



The worst
(except airplanes)




Sources of Transportation Air Pollution
Smog and soot → Health and welfare impacts
CO_2 and other greenhouse gases → Climate change
Light Duty Vehicles
Heavy Duty Vehicles
Motorcycles
Aircraft
Nonroad Diesel Equipment
Marine Engines/ Ocean Vessels
Locomotives
Lawn & Garden
Recreation Vehicles
ON ROAD
NONROAD
Solutions for Transportation Air Pollution
Emission reductions → Cleaner air & better health
Catalytic converters in conjunction with unleaded gasoline and low sulfur levels significantly reduce hydrocarbon & nitrogen oxide emissions
Fuel standards reduce exposure to pollutants like lead and benzene
Renewable fuels reduce CO_2 emissions
Engine technologies like computer controls, variable valve timing, multi-valve engines, turbo charging & gasoline direct injection improve fuel economy & reduce CO_2 emissions
Transmission technologies like 7+ speeds, dual clutch transmissions (DCTs), & continuously variable transmissions (CVTs) improve fuel economy & reduce CO_2 emissions
Diesel filters reduce particulate matter from on road & off road diesel engines
Alternative vehicle technologies like plug-in electric vehicles & fuel cells = zero tailpipe emissions
Better transportation planning for passengers & freight reduce emissions & fuel use
EPA United States Environmental Protection Agency

Our target

Why Soft Computing for Route Planning?

- **Limitations of Classical Methods**
  - Urban route planning is NP-hard; exact mathematical models fail to scale as network size and constraints grow.
  - Deterministic models struggle with dynamic urban changes (accidents, peak-hour shifts).
- **Handling Real-World Complexity & Uncertainty**
  - Traffic networks are inherently non-deterministic and ill-defined.
  - Soft Computing comfortably incorporates **fuzzy traffic data** and imprecise human/environmental factors.
- **The Bio-Inspired Advantage**
  - Combines global exploration (evolutionary concepts) with local exploitation (memetic refinement).
  - Finds high-quality, **near-optimal routes within practical time limits** (avoiding exhaustive search).

Introduction to Evolutionary & Bio-Inspired Algorithms

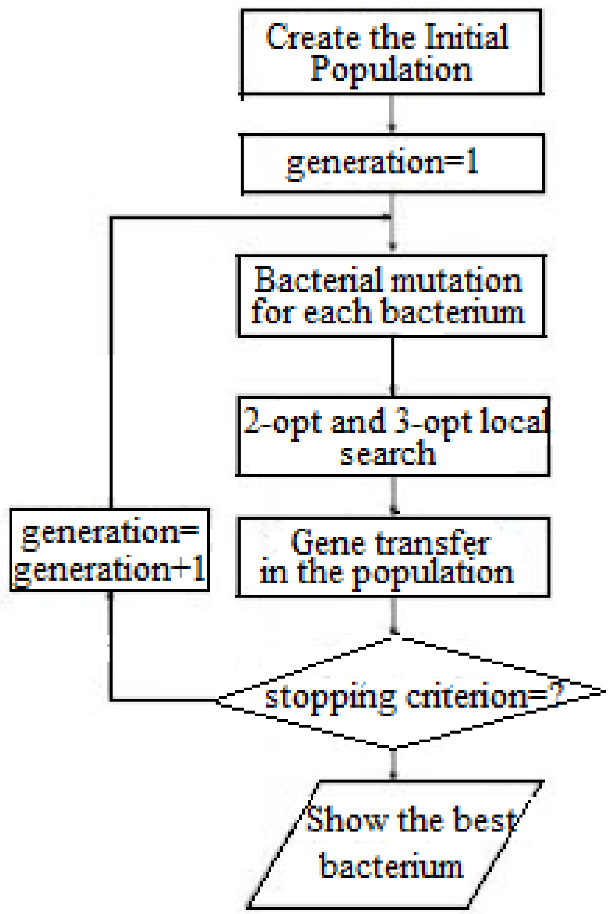
- **Natural Selection in Route Optimization**
  - Inspired by biological evolution: survival of the fittest.
  - A population of candidate routes evolves over generations to find the most efficient path.
- **Core Mechanisms (High-Level)**
  - **Representation:** Each "individual" (chromosome) represents a complete vehicle route.
  - **Fitness Function:** Evaluates route quality (minimizing travel time, fuel, CO_2, delays, and penalties).
  - **Variation (Operators):** Combining and modifying existing routes to discover better paths.
- **Key Advantage over Classical Search**
  - Avoids getting trapped in local optima (e.g., suboptimal traffic routes).
  - Evaluates multiple route alternatives simultaneously in complex urban networks.

Memetic algorithms

- Memetic algorithms: evolutionary based method + nested local search → eliminates the weaknesses of both methods
  - Evolutionary based methods:
    - search in the global search space
    - slow convergence speed
  - Local search techniques:
    - search in the neighborhood of the current solution
    - converge to the nearest local optimum

Discrete Bacterial Memetic Evolutionary Algorithm (DBMEA)

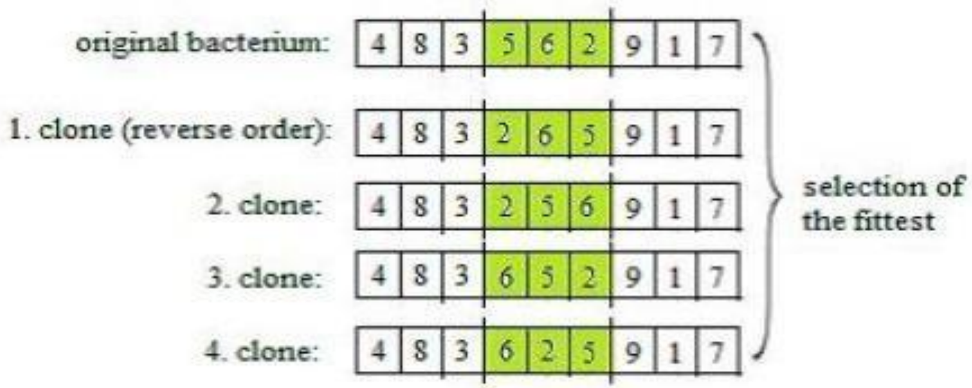
- **Biological Inspiration**
  - Mimics the evolutionary behaviour and genetic adaptation of bacterial populations.
- **Two-Level Hybrid Architecture (Memetic Approach)**
  - **Global Search (Bacterial Evolution):** Bacterial mutation and gene transfer
  - **Local Search (Memetic Refinement):** 2-opt and 3-opt
- **Key Strengths for Combinatorial Optimization**
  - Fast convergence while maintaining high population diversity (prevents getting stuck in local optima).
  - Highly effective for discrete, NP-hard problems like TSP, Time-Dependent TSP, and vehicle routing.
  - Easily adaptable to complex real-world constraints (time windows, congestion zones, and travel times).



Discrete Bacterial Memetic Evolutionary Algorithm (DBMEA)

Bacterial mutation

- optimizes (or does not…) the bacteria individually
  - Cut the bacteria into fixed length segments
  - Create N_{clones} clones from the original
  - Carry out the following process until all segments are mutated:
    1. choose randomly a segment
    2. modify randomly the selected segment in the clones
    3. select the fittest from the original and the clones
    4. copy the segment of the fittest individual to the others



Discrete Bacterial Memetic Evolutionary Algorithm (DBMEA)

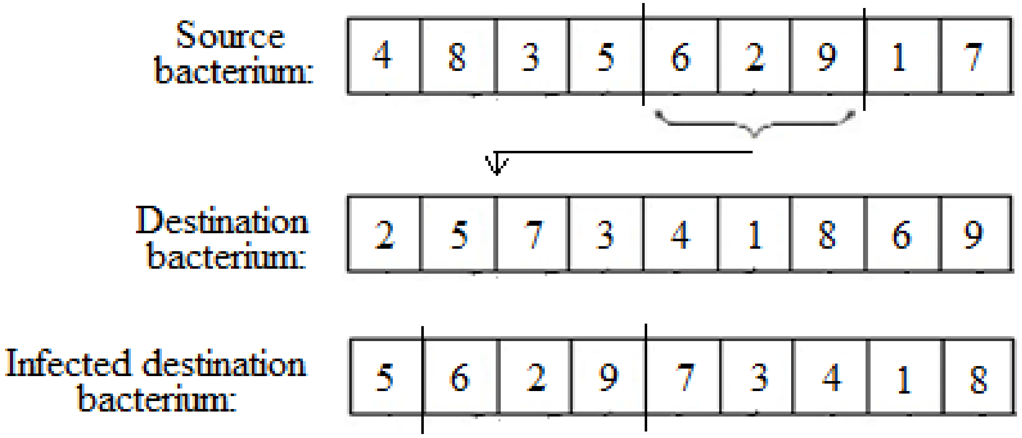
Gene transfer

allows the transfer of information between the bacteria in the population

1. sort the population in a descending order according to their fitness values and divided into two parts (superior („good” bacteria) and inferior („bad” bacteria) half)

2. choose a bacterium from both halves randomly

3. copy a randomly selected fixed length part of the selected „good bacterium into the selected „bad” bacterium



Computationally Challenging Routing Problems targeted

Examined Problems in Urban Logistics

- Traveling Salesperson Problem (**TSP**) & TSP with Time Windows (**TSPTW**)
- Time-Dependent TSP (**TDTSP**) & Traveling Repairman Problem (**TRP**)

The Computational Challenge (NP-Hard Class)

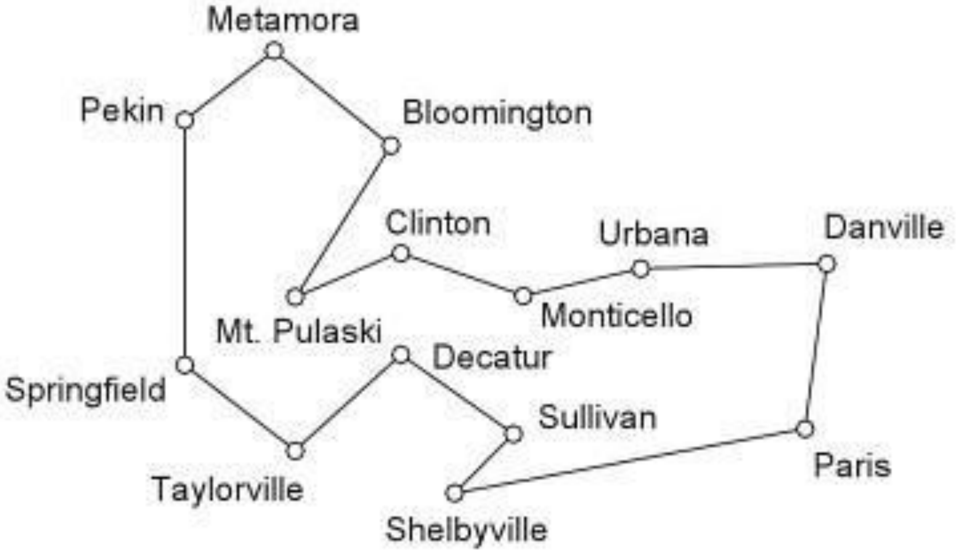
- **Combinatorial Explosion:** Route possibilities grow exponentially with each added stop.
- **Exact Solver Limits:** Classical mathematical algorithms fail to scale for large urban networks within practical time limits.

Practical Solution: Bio-Inspired Metaheuristics

- Delivers high-quality, operational routes in **realistic timeframes**.

The Traveling Salesman Problem

- A Salesman travels around a given set of cities, and returns to the beginning, covering the smallest total distance
- Several application areas (logistics, microchip manufacturing etc.)



Comparison of minimal tour lengths TSP

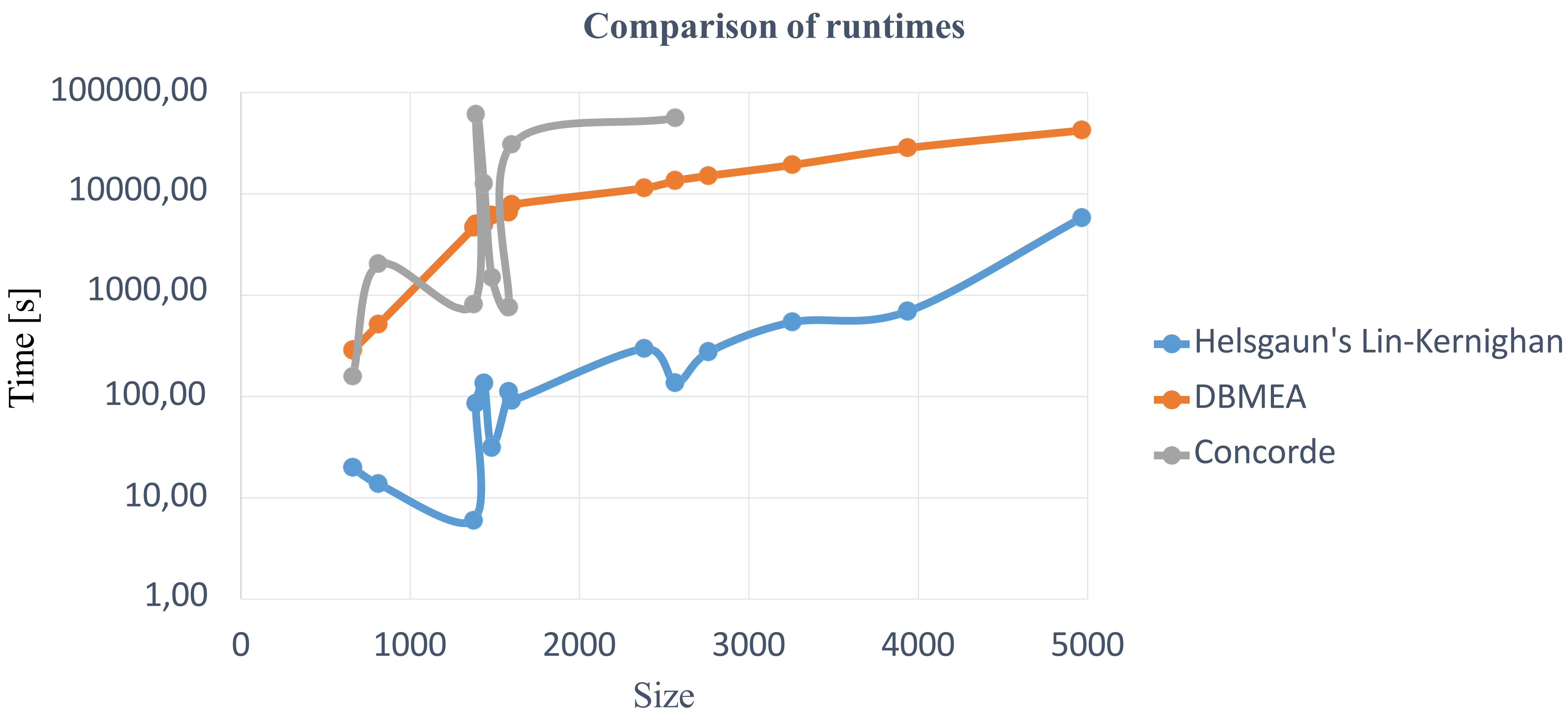
Kóczy, L. T., Földesi, P., & Tüű-Szabó, B. (2018). Enhanced discrete bacterial memetic evolutionary algorithm-an efficacious metaheuristic for the traveling salesman optimization. *Information Sciences*, *460*, 389-400.

Instance	DBMEA		Helgaun's Lin-Kernighan		Concorde	gap DBMEA [%]
	best value	avg. Value	best value	avg. Value	optimal value	
xql662	2 550.84	2 550.84	2 550.84	2 550.92	2 550.84	**0**
dkg813	3 243.41	3 243.41	3 243.41	3 243.68	3 243.41	**0**
dka1376	4 738.61	4 743.20	4 736.87	4 736.87	4 736.87	**0.04**
dca1389	5 156.43	5 158.06	5 152.69	5 153.08	5 152.69	**0.07**
dja1436	5 332.19	5 335.93	5 328.21	5 329.01	5 328.21	**0.07**
icw1483	4 467.22	4 472.12	4 465.88	4 465.94	4 465.88	**0.03**
rbv1583	5 446.09	5 449.99	5 444.67	5 444.91	5 444.67	**0.03**

Comparison of minimal tour lengths TSP

Instance	DBMEA		Helgaun's Lin-Kernighan		Concorde	gap DBMEA [%]
	best value	avg. Value	best value	avg. Value	optimal value	
rby1599	5 589.13	5 592.02	5 584.16	5 585.20	5 584.16	**0.09**
dea2382	8 154.34	8 160.93	8 146.94	8 148.87	-	**0.09**
pds2566	7 774.91	7 781.57	7 766.71	7 766.89	7 766.71	**0.1**
bch2762	8 376.18	8 380.63	8 368.56	8 370.33	-	**0.09**
fdp3256	10 104.36	10 108.69	10 091.29	10 092.15	-	**0.13**
dkc3938	12 715.99	12 723.46	12 704.22	12 704.31	-	**0.09**
xqd4966	15 554.91	15 560.49	15 530.82	15 531.00	-	**0.16**

Comparison of run times TSP



Time Dependent Traveling Salesman Problem (TDTSP)

- An extension of the TSP
- NP-hard problem
- The costs between nodes vary in time (two values)
- More applicable in the real life than the TSP (transportation, logistics) The aim is to find the tour with the lowest cost which visits each node once

Time Dependent TSP

- Schneider (Simulated Annealing method): bier127 reference problem with different jam factors
- Traffic jam region (coordinate of the left corner point: (7080, 7200), width: 6920, height: 9490)
- Working hours of the salesman: between 9 am and 3 pm. Rush hour: from 12 pm to 3pm
- In this period the cost is proportional with the original cost and with the traffic jam factor
- Travel speed is computed by dividing the total distance of the optimal TSP tour (118293.524) by the working hours of the salesman
- Travel speed is held constant for all values of the traffic jam factor

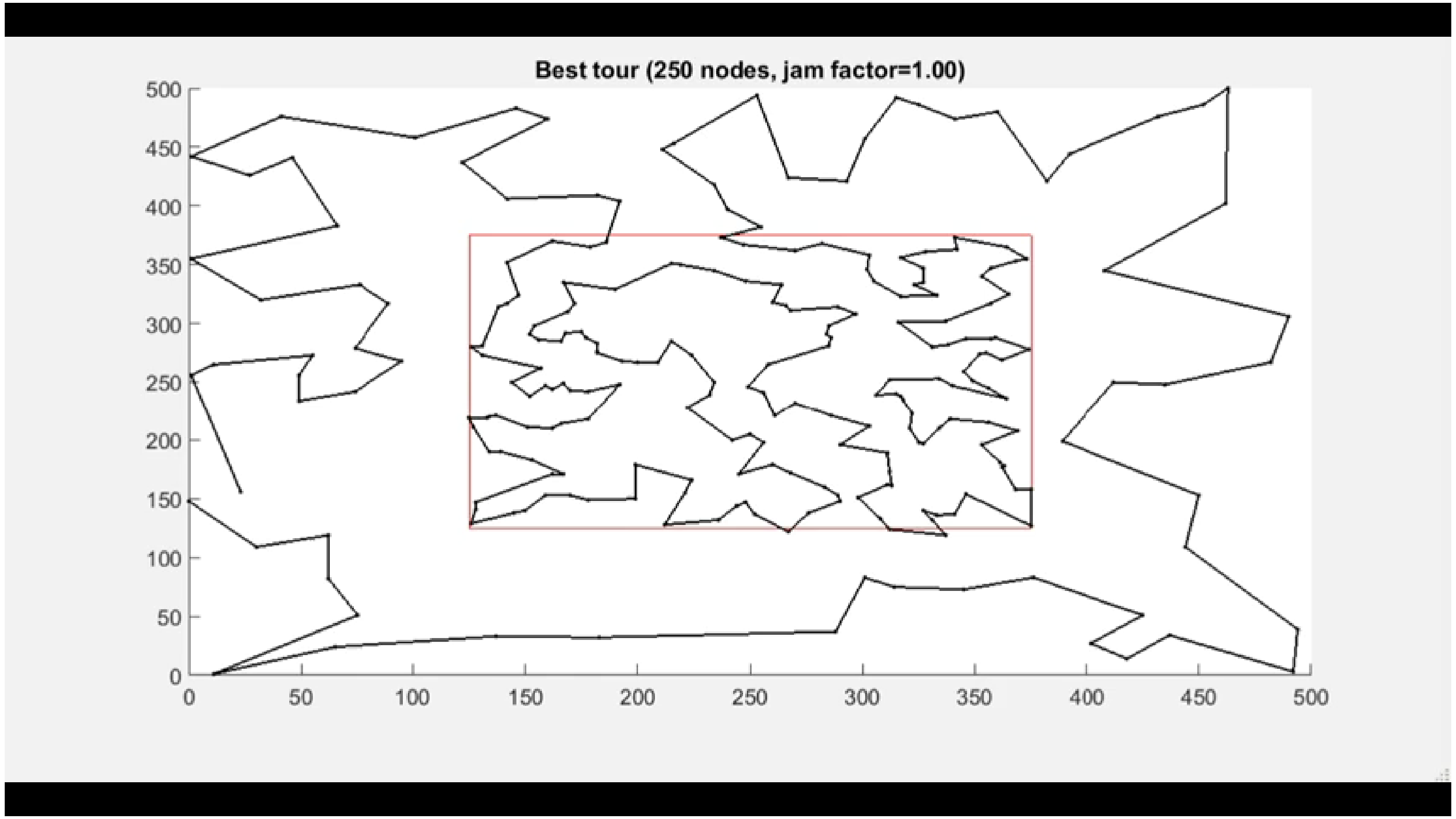
Computational results (TDTSP)

Two self-generated problems with 250 nodes:

- A traffic jam region is defined (coordinate of the left corner point (125, 125), width is 250, and the height is 250)
- Vertex coordinates have been generated by a uniform random generator, 70% (s250_1) and 50% (s250_2) of the nodes were placed within the traffic jam region
- The working hours and the calculation of travel speed is the same as in the case of bier127

Best tours with different traffic jam factors (s250_1)

Tüű-Szabó, B., Földesi, P., & Kóczy, L. T. (2018, May). Discrete bacterial memetic evolutionary algorithm for the time dependent traveling salesman problem. In *International Conference on Information Processing and Management of Uncertainty in Knowledge-Based Systems* (pp. 523-533). Cham: Springer International Publishing.



The Triple Fuzzy Time Dependent TSP (3FTD TSP)

The Weak Point of the Classic TD TSP

- TD TSP is able to achieve more adequate results in determining the overall cost under real life conditions, but a weak point of the model is that a single concrete (crisp) number is used to represent both the proportional jam factor for the whole traffic jam regions, and the rush hour time period.
- Here, the increased cost (in the jam region) is calculated from the basic cost assigned to the edges multiplied by the (crisp) jam factor (only in the rush hour period).
- In reality, there are many uncertain elements influencing the real costs and these are taken into consideration in the novel fuzzified approach

Triple Fuzzy TD TSP (3FTD TSP)

- Various types of uncertainties are not taken into consideration in the original TD TSP
- The Triple Fuzzy Time Dependent Traveling Salesman Problem (3FTD TSP) is a realistic approach, thoroughly extended and fuzzified, but based on the original TD TSP model.
- In the 3FTD TSP, fuzzy values are applied
  - (1) for the costs between pairs of nodes
  - (2) for defining the geographical area(s) of the traffic jam region(s), and also
  - (3) for defining the rush hour period(s).

Koczy, L. T., Foldesi, P., Tuu-Szabo, B., & Almahasneh, R. (2019, June). Modeling of fuzzy rule-base algorithm for the time dependent traveling salesman problem. In *2019 IEEE International Conference on Fuzzy Systems (FUZZ-IEEE)* (pp. 1-6). IEEE.

Example for The 3FTD TSP Model

In the next simple example, the total cost of the tour will be calculated as follows.

The costs are represented by asymmetrical triangular fuzzy numbers. Each fuzzy cost has the form

$$C'(t) = (C'_L, C'_C, C'_R) = (C_L, C_C, C_R) \times (1 + (jam_{factor} - 1) \times \mu_1 \times \mu_2)$$

- Where μ1 and μ2 are calculated based on the membership functions of the Fuzzy Jam Region J and of the Fuzzy Traffic Rush Hour R.
- Everywhere, piecewise linear functions are used

Example for The 3FTD TSP Model

A valid solution for the problem is a permutation of the nodes:

$$p_1, p_2, \cdots, p_n, p_1$$

where p1 is a fixed node (the starting and ending node) of the traveler.

The time needed for visiting the first node from the starting node is

$$t_{P_1,P_2} = \frac{COG(C'_{P_1,P_2}(t=0))}{v}$$

Example for The 3FTD TSP Model

The time dependency of the cost matrix is represented by a virtual distance value.

If a cost element is growing then the required time is growing as well, since v is considered to be constant. The cost between Pk and Pk+1 is

$$t_{elapsed_k} = \sum_{i=1}^{k-1} t_{P_i,P_{i+1}}$$

$$t_{P_k,P_{k+1}} = \frac{COG(C'_{P_k,P_{k+1}}(t_{elapsed_k}))}{v}.$$

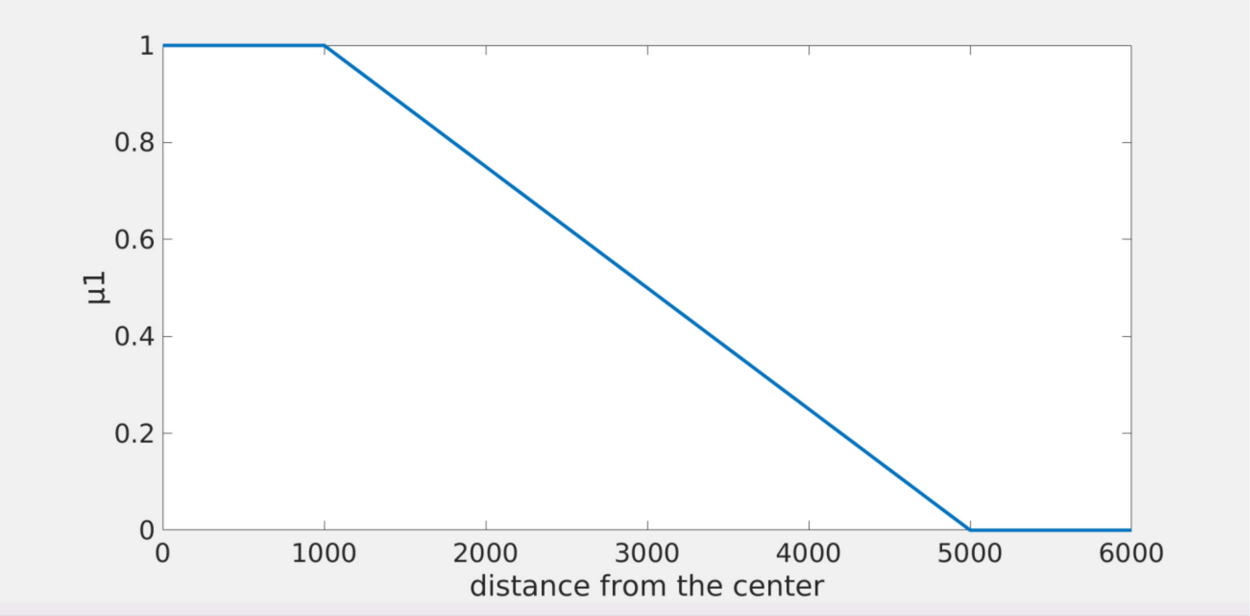
The total cost will be:

$$S = \sum_{i=1}^{n-1} COG\left(C'_{P_i,P_{i+1}}\left(t_{elapsed_i}\right)\right) + COG\left(C'_{P_n,P_1}\left(t_{elapsed_n}\right)\right)$$

Example for The 3FTD TSP Model

The Membership Function of the Jam Regions

The costs between nodes in the traffic jam region are higher when in the rush hour period. The degrees are the highest in the „city center" area.



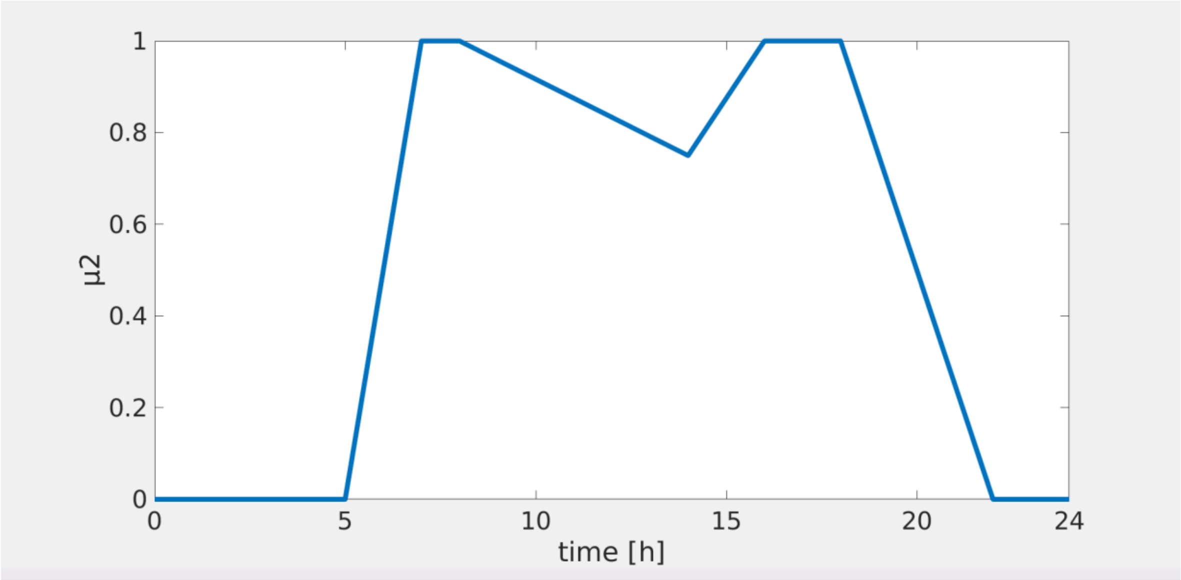
Example for The 3FTD TSP Model

Membership Function R of the Rush Hour Period

In this Example, there are two rush hour peaks, from 7 a.m. to 8 a.m. and from 4 p.m. to 6 p.m. (Between the two peaks the traffic is lower, but also higher than in the night, when it is considered to be minimal.)

The breakpoints of the membership function are

[0, 5, 7, 8, 14, 16, 18, 22, 24], and their respective membership values are: [0, 0, 1, 1, 0.75, 1, 1, 0, 0].



Example for The 3FTD TSP Model

Computational Results for the 3FTD TSP example

The obtained results are validated in the context of the known optimum solution for the original TD TSP, and the solutions by Schneider and our own results for the TD TSP

TABLE I. COMPUTATIONAL RESULTS FOR 3FTD TSP

jam factor	DBMEA best elapsed time	DBMEA average elapsed time	DBMEA average runtime [s]
1.00	19.322	19.499	88.324
1.05	19.583	19.607	90.138
1.20	19.666	19.742	91.631
1.50	20.179	20.291	88.790
2.00	20.847	20.912	81.111
3.00	21.562	21.633	79.027
5.00	22.037	22.196	84.965
10.00	22.969	23.102	78.250

Example for The 3FTD TSP Model

Computational Results for the 3FTD TSP example

The results indicate that the DBMEA was able to find **high quality solutions** for the 3FTD TSP problem, with reasonable membership functions J and R, even with higher jam factors, for the proposed new problem.
With high jam factors, the **tour lengths were** obviously **longer** for the 3FTD TSP problem, because then the traffic jam period is longer compared to the TD TSP

TABLE II. COMPARISON OF THE RESULTS FOR THE 3FTD TSP AND THE TD TSP

jam factor	DBMEA 3FTD TSP best value	TD TSP best value
1.00	115929.0	118293.5
1.05	117498.6	119053.2
1.20	117997.2	119714.1
1.50	121076.4	120571.7
2.00	125083.8	121125.2
3.00	129369.0	121125.2
5.00	132220.2	121125.2
10.00	137813.4	121125.2
20.00	139286.4	121125.2
50.00	144363.1	121125.2
100.00	147520.2	121125.2

Conclusions for 3FTD TSP

- 3FTD TSP is a novel approach that models real life situation for the TD TSP.
- It assumes imprecise (fuzzy) values representing the uncertainties associated with the geographic extent of the traffic jam area, the occurrence of the rush hour phenomenon in time, and the edge costs themselves, which may depend on several unknown or non-deterministic factors.
- These uncertainties were represented by piecewise linear membership function fuzzy sets (except in the case of the rush hour period, convex and normal fuzzy sets) without loss of generality. This way the model is capable to express the uncertain costs influenced by the jam situations, and ultimately, to calculate the overall tour length quantitatively.
- The results obtained are in accordance with expectations based on similar, related simulation results.

Intuitionistic Fuzzy Time Dependent Traveling Salesman Problem (IFTD TSP)

- **Background (Why is it needed?):**
  - Traditional and standard fuzzy models handle uncertainty, but they fail to separate membership and non-membership degrees independently.
  - In real-world traffic scenarios (e.g., congestion, peak hours, tolls, travel times), data is not only imprecise, but can also be incomplete or contradictory.
- **Core of the Model:**
  - Uses **Intuitionistic Fuzzy Sets (IFS)** to calculate the jam region and rush hour period cost factors.
  - Applies three main metrics:
    - **Membership degree (μ):** To what extent a **road segment** belongs to the Jam Region (J), or a **given time** belongs to the Rush Hour (R).
    - **Non-membership degree (ν):** To what extent it is certain that the segment is **outside** the jam region, or the time is **outside** the rush hour.
    - **Hesitation degree ($\pi = 1 - \mu - \nu$):** The measure of missing information, unpredictable traffic, or boundary uncertainty for both spatial and temporal factors.

Why Intuitionistic Fuzzy Logic? (Practical Impact)

The Core Advantage: Risk-Aware Decision Making

- Traditional models only look at average travel times. The Intuitionistic Fuzzy (IF) model introduces a **Risk Premium** based on the hesitation degree (π).

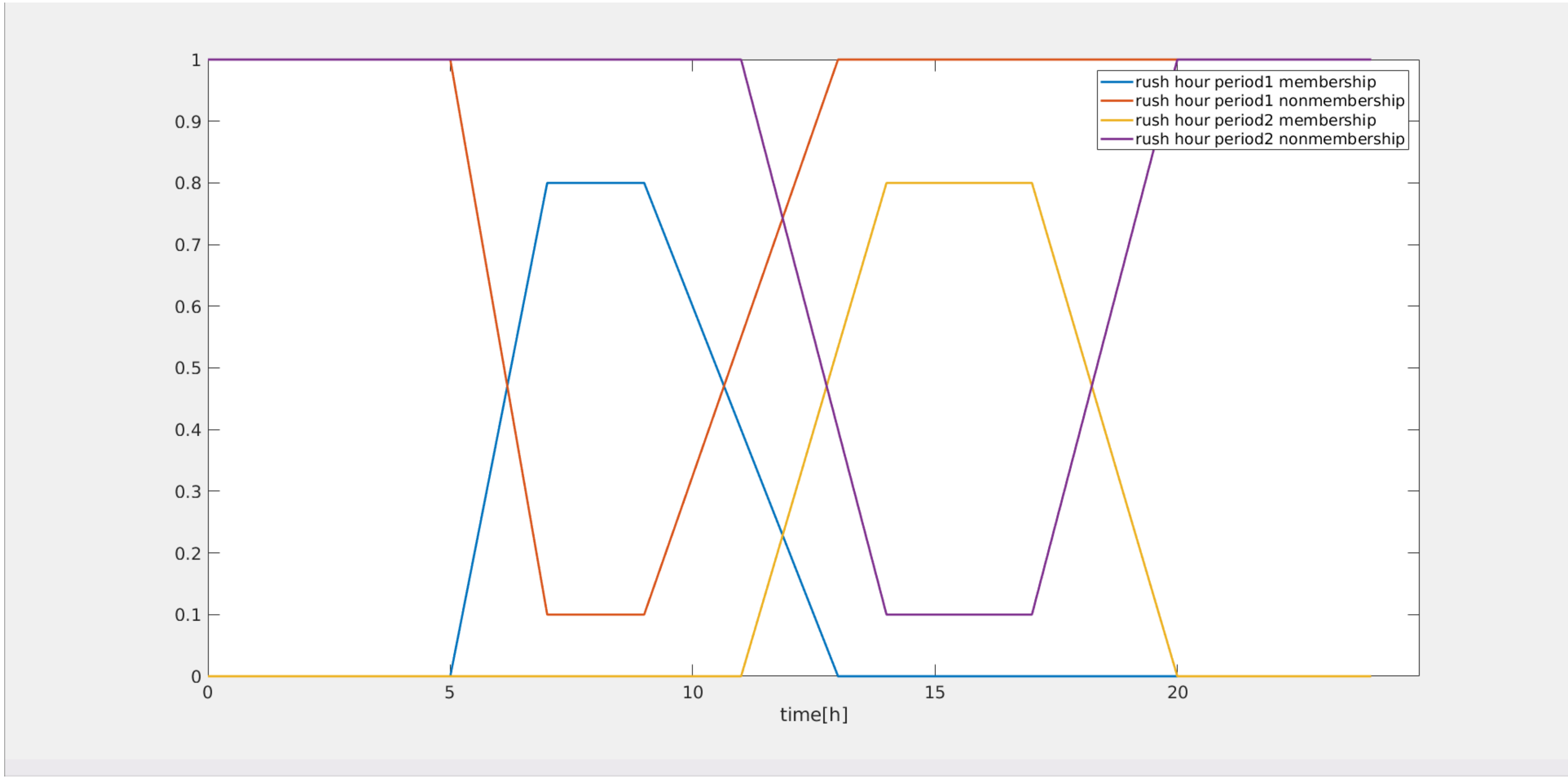
Quantifying Unpredictability (π):

- High hesitation (π) represents unpredictable traffic (e.g., frequent accidents, unstable traffic lights, volatile rush hours).

Risk vs. Distance Trade-off:

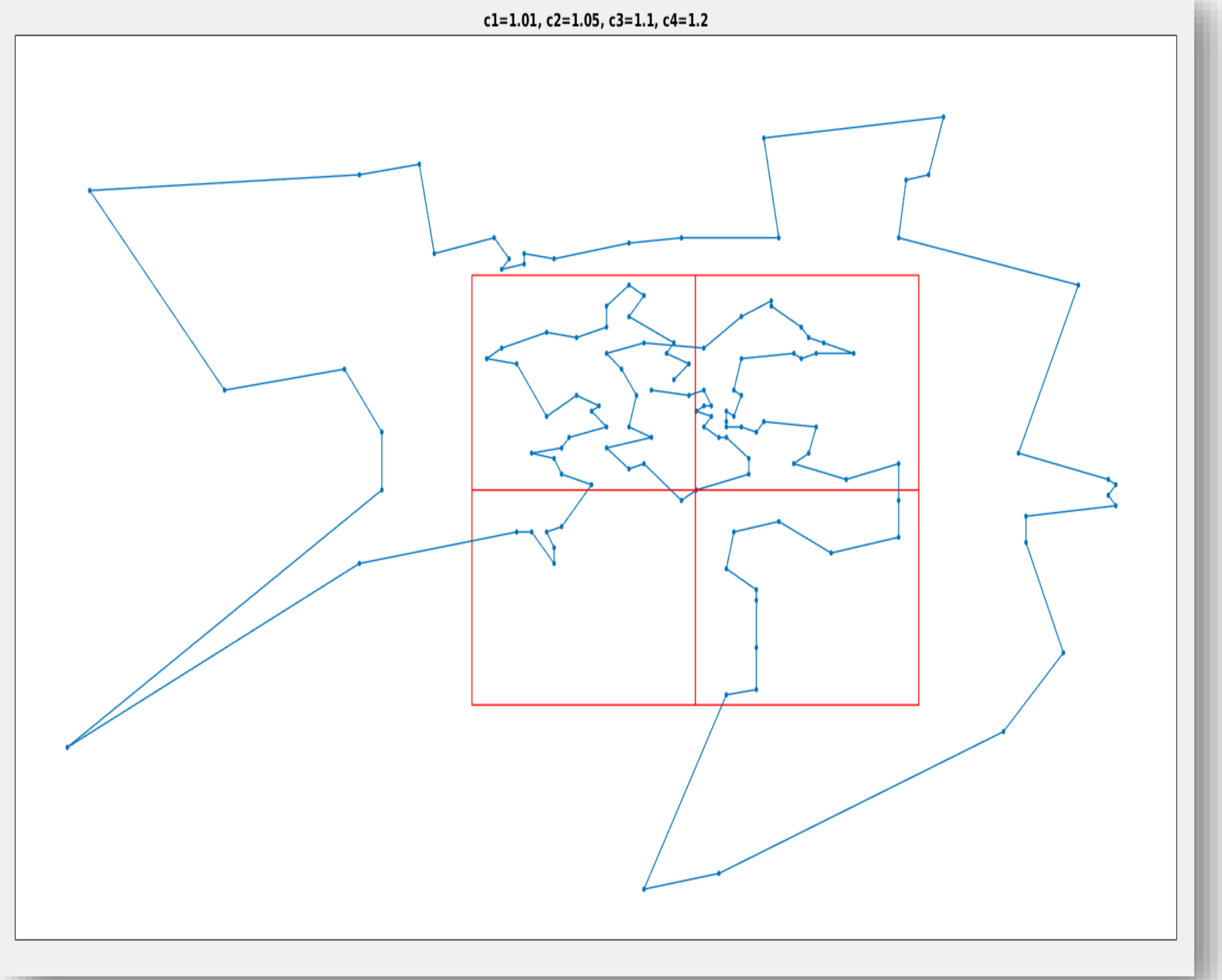
- A route with high hesitation is penalised in the cost function. As a result, the algorithm will **penalise high-uncertainty routes** even if they appear shorter on paper.

The fuzzy membership and non-membership functions



Fuzzy membership and non-membership functions of the rush hour periods

IFTD TSP using DBMEA

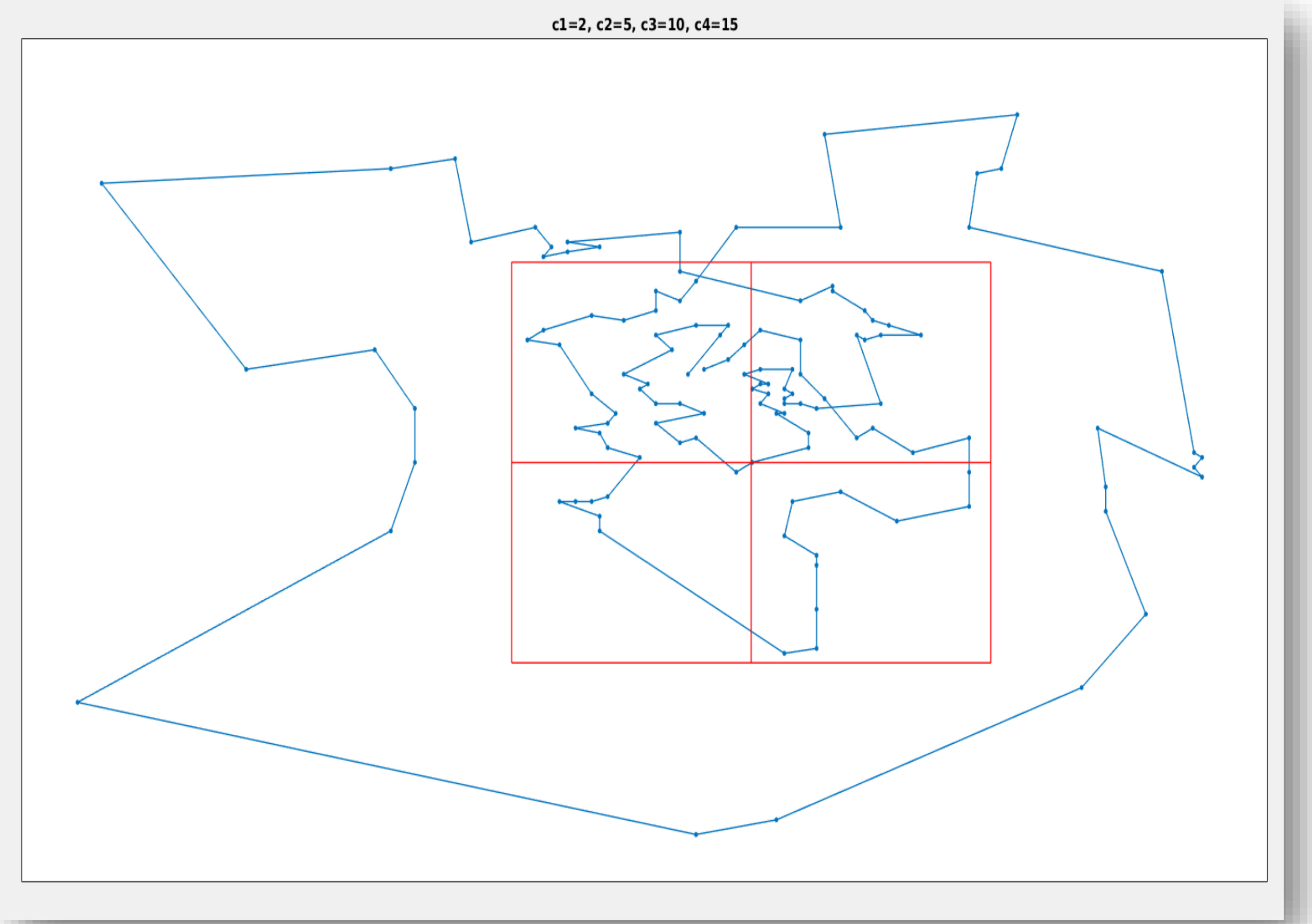


- The jam region was divided into four equal sized subdivisions (see red rectangles)
- The best tour for cost factor values

$$(c_1 = 1.01\ c_2 = 1.05\ c_3 = 1.1\ c_4 = 1.2)$$

Almahasneh, R., Tüű-Szabó, B., Földesi, K., & Kóczy, L. T. (2019, June). Intuitionistic Fuzzy Model of Jam Regions and Rush Hours for the Time Dependent Traveling salesman Problem. In *Proceedings of the IFSA World Congress and NAFIPS Annual Conference*.

IFTD TSP using DBMEA



- The best tour for cost factor values

$(c_1 = 2\ c_2 = 5\ c_3 = 10\ c_4 = 15)$

Conclusions

1. Methodological & Algorithmic Contributions

- **Advanced Soft Computing Framework:** Successfully developed and applied the **Discrete Bacterial Memetic Evolutionary Algorithm (DBMEA)**, combining bacterial global exploration with fast local search operators (2-opt / 3-opt).
- **High Efficiency & Scalability:** Demonstrated near-optimal solutions with significantly lower computational execution times compared to exact solvers (e.g., Concorde) on standard TSPLIB benchmarks.

2. Uncertainty Modelling

- **From Deterministic to Intuitionistic Fuzzy:** Advanced route optimization from basic TSP to **Time-Dependent TSP (TDTSP)**, **3-Factor Fuzzy TDTSP (3FTD TSP)**, and ultimately **Intuitionistic Fuzzy TDTSP (IFTD TSP)**.
- **Risk-Aware Decision Making:** Incorporated hesitation degrees (π) to account for uncertainty and risk in travel time estimation.

SZÉCHENYI
EGYETEM
UNIVERSITY OF GYŐR

Thank you for your attention!